

Scale Buys Interpolation, Structure Buys a Horizon: Certified Predictability for Equivariant World Models

How far a world model can promise — tightly, and only with structure.

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Abstract

Scale buys interpolation; structure buys a certified horizon. A world model’s average error says nothing about whether a *particular* prediction can be trusted, or for how long. For **equivariant** latent world models we give a computable, multi-step **certificate of the predictable horizon**: T -step rollout error is provably constant over each symmetry orbit (Theorem A) and stratified channel-by-channel by the predictor’s Lyapunov spectrum, $T_j(\epsilon) \sim \log(1/\epsilon)/\lambda_j$. The horizon is **two-sided** — a matching lower bound makes approximate equivariance provably horizon-limited (Proposition 6) — and the certificate is **exclusive to structure**: orbit-constant error *characterizes* equivariance (Lemma 2), so no non-equivariant model has it at any scale. Empirically, on 40-D Lorenz-96 only a $\mathbb{Z}_N$ -equivariant network recovers the full Lyapunov spectrum ($R^2=0.98$); identically-trained dense and recurrent baselines fail.

Because the spectrum is faithful, the certificate **acts, a priori**: under a fixed sensing budget a $c\times$ -inflated certificate provably needs $c\times$ the budget (Proposition 9), and the equivariant certificate meets a budget its inflated dense counterpart cannot — with zero calibration data. The same read-out, unchanged, **audits public pretrained world models training-free**: official TD-MPC2 checkpoints land on the certificate’s own scope taxonomy — calibrated where strongly expansive (ratio 0.94–1.02), optimistic where weakly expansive, correctly abstaining where contracting — and a deployed sensing-budget monitor then replicates that map **cell-by-cell, out-of-sample, at zero new estimation**. Across the official 1M–317M multitask ladder, calibration does **not** improve with parameters: scale buys interpolation, not a calibrated horizon.

1. Introduction

A world model is judged by its average prediction error. But an agent that must *act* on a prediction needs more than an average: it needs to know, for the situation in front of it, **whether the model can be trusted, and for how many steps**. Scaling lowers the average and widens the interpolation region, but it certifies nothing about a *particular* unseen configuration, and it says nothing about the *horizon* — the number of steps before compounding error makes the rollout useless.

Equivariance gives *some* per-situation guarantee: a network commuting with a group G behaves identically across each orbit — used to certify adversarial robustness and tighten conformal sets. But every such result is **single-shot**: one input, one prediction. A world model predicts a *trajectory*, and the question that matters for planning — *how far ahead is the guarantee valid?* — has no answer in that literature.

This paper answers it. We show that an equivariant latent world model admits a **certificate of its predictable horizon**, stratified channel-by-channel by the predictor’s spectrum, and we prove the horizon is **tight**. The contributions are:

1. **A tight certified horizon (§3.2)**. For an equivariant model the T -step rollout error is orbit-constant (Theorem A); under approximate equivariance, channel j is certified to $T_j(\epsilon) \sim \log(1/\epsilon)/\lambda_j$ (Theorem B). The central new result is a **matching lower bound** (Proposition 6): an ϵ -approximately-

equivariant model — even with a *perfect* predictor — has orbit-error-variation exactly $\epsilon e^{\lambda T}$, so the horizon is $\Theta(\log(1/\epsilon)/\lambda)$, **no better**: *approximate equivariance is horizon-limited; only exactness or conservation reaches an unbounded horizon*. A scope theorem (Proposition 7) says *when* the measured spectrum governs a *learned* model’s horizon, a continuity bound (Proposition 8) makes the lift rigorous, and the staircase lifts to learned models of genuinely chaotic systems (E2: exponents matched to 1–12%), the near-neutral PushT interior being the predicted degenerate case.

2. **The certificate is exclusive to structure (§3.1).** The converse (Lemma 2): orbit-constant error against every equivariant target $\iff$ equivariance — **no non-equivariant model has the certificate at any size**, an architectural impossibility, proved, not asserted.
3. **The Noether hinge (§3.3).** The unbounded-horizon channels are *exactly* the conserved/invariant ones: Proposition 4 forces each charge’s isotypic placement; Proposition 5 turns conservation into a long-horizon guarantee (T -step charge error $\leq T\eta$, linear) — the slow, certifiable subspace is the group-invariant one.
4. **Single-shot certified equivariance is our $T=1$ slice (§5)** — the one-step, resolution-free special case; the multi-step picture is validated on a contact simulator, non-abelian $\text{SO}(3)$, and raw pixels (frame averaging keeping the certificate *accuracy-neutral*).

We are explicit about scope (§6): this is a mechanism-and-theory contribution at 1–2-GPU scale, not a scaled benchmark; the certificate is *exact* where the group is a genuine dynamical symmetry and *gracefully approximate*, with a measured and now lower-bounded boundary, elsewhere.

One certificate, a universal half and an exclusive half. The spectral audit (Theorem B’s law) applies to **any** C^1 latent loop — that is how E13–E15 audit non-equivariant public models; what is **exclusive to structure** is *a-priori trust in the audited number itself* (Lemma 2; E2: a dense spectrum can be silently wrong at one-step $\text{relMSE } 10^{-5}$). A generic model buys that trust with held-out divergence data; an equivariant model has it from the Jacobian alone. §4’s closing paragraph returns to this division.

What is new. A prior-art sweep finds the corner *empty*: no work combines equivariance with a certified, two-sided spectral horizon; the closest guarantee-bearing neighbours each miss the intersection (Conradie et al. 2026; Lillemark et al. 2026; Mo 2026) (§5). The new results: **(i)** Proposition 6’s matching lower bound — growth seeded by the *equivariance residual* rather than an initial-condition error — making approximate equivariance provably horizon-limited; **(ii)** the scope/continuity pair (Props. 7–8) lifting the law to *learned* chaotic models, where shadowing cannot transfer the exponent; **(iii)** the structure-vs-recurrence separation at 40-D (E2), with the recurrent failure forced by the conditional-Lyapunov condition; **(iv)** Theorem B turning the certificate from *measured* into *literal* — sound a priori, tight on uniform hyperbolicity, self-abstaining elsewhere; **(v)** actionability **a priori, with zero calibration data** (E12; Proposition 9); and **(vi)** a training-free trustworthiness audit of public pretrained world models (E13–E14) that a deployed monitor replicates **cell-by-cell, out-of-sample** (E15), its decision boundary formalized as a regret decomposition (Proposition 11). These assemble (Algorithm 1) into one certificate *exclusive to structure* (Lemma 2) whose unbounded-horizon subspace is the conserved/invariant one (the Noether hinge). We credit the classical pillars we build on — the Lyapunov/NWP horizon law, invariant decision theory, and the $e^{\lambda T}$ growth — explicitly in §3 and §5.

2. Setup and assumptions

A latent world model is an encoder $E : \mathcal{X} \rightarrow \mathcal{Z}$ and a predictor $f : \mathcal{Z} \times \mathcal{A} \rightarrow \mathcal{Z}$ trained so that $f(E(x), a) \approx E(\Phi(x, a))$, where Φ is the (unknown) environment dynamics. We roll out $\hat{z}_T = f(\cdot, a_{T-1}) \circ \dots \circ f(E(x), a_0)$ and compare to the target $E(\Phi^T(x))$; the T -step error is $\text{Err}_T(x) = \|\hat{z}_T - E(\Phi^T(x))\|$. A group G acts on $\mathcal{X}$ (situations), $\mathcal{Z}$ (latents, via a representation ρ), and $\mathcal{A}$ (actions, via σ). We use:

- **(A1) Encoder equivariance:** $E(g \cdot x) = \rho(g) E(x)$.
- **(A2) Predictor equivariance:** $f(\rho(g)z, \sigma(g)a) = \rho(g) f(z, a)$.
- **(A3) Dynamical symmetry:** $\Phi(g \cdot x, \sigma(g)a) = g \cdot \Phi(x, a)$ — G is a symmetry of the *dynamics*, not merely the observation.
- **(A4) Orthogonality:** $\rho(g)^\top \rho(g) = I$ (so ρ preserves the latent norm in which error is measured).

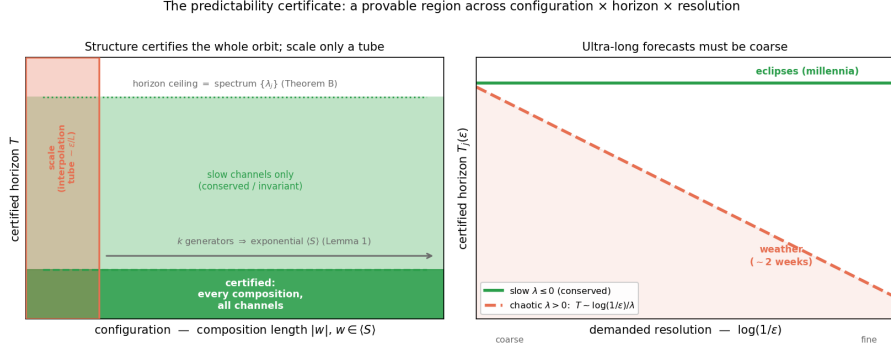


Figure 1: The certificate at a glance. **Left:** an equivariant model certifies the *entire* generated monoid $\langle S \rangle$ from k generator checks (Lemma 1), up to a horizon ceiling set by the predictor spectrum (Theorem B, tight by Proposition 6); a non-equivariant model of any size certifies only an interpolation tube. **Right:** the horizon-resolution law $T_j(\epsilon) \sim \log(1/\epsilon)/\lambda_j$ — conserved/invariant channels ($\lambda \leq 0$) are certified to all horizons, chaotic ones shrink with demanded resolution.

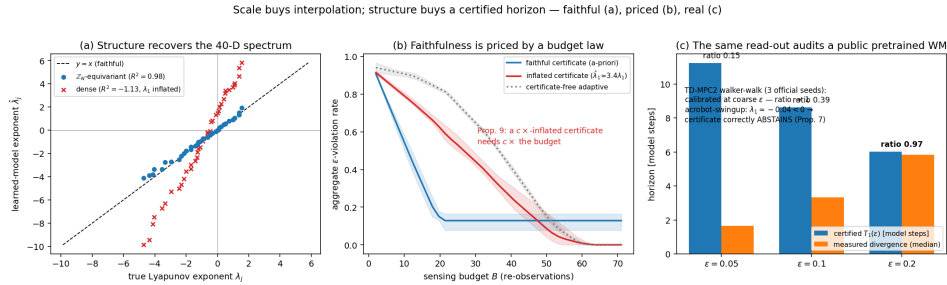


Figure 2: **The paper in one figure.** (a) **Faithful:** on 40-D Lorenz-96 the $\mathbb{Z}_N$ -equivariant model recovers the full Lyapunov spectrum ($R^2=0.98$) where an identically-trained dense model's is garbage ($R^2<0$) — E2. (b) **Priced:** under a fixed sensing budget the faithful certificate meets the budget; the inflated one over-observes and starves it (Proposition 9) — E12. (c) **Real:** the same training-free read-out audits official TD-MPC2 checkpoints — calibrated where expansive, abstaining where contracting — E13.

- **(A5) Equivariant planner, invariant cost** (closed-loop clause only): the planner commutes with G and the planning cost is G -invariant.

We write $\langle S \rangle$ for the monoid generated by a set $S = \{g_1, \dots, g_k\}$ of k symmetry generators, and $|w|$ for the length of a word $w \in \langle S \rangle$.

3. The certificate

3.1 The configuration axis, and why only structure has it

Theorem A (orbit-constant error). Under (A1)–(A4), for every $w \in \langle S \rangle$, all x , and any action sequence (transported by w), $\text{Err}_T(w \cdot x) = \text{Err}_T(x)$ (action-explicit statement and proof in Appendix B). *Proof sketch.* The rolled-out predictor is equivariant (Lemma 1), so prediction and target transport by the same $\rho(w)$; subtract and use (A4). $\square$

Lemma 1 (composition closure). If (A1)–(A3) hold on each generator $g_i \in S$, they hold on every word $w \in \langle S \rangle$ ((A4) is automatic for all w). Thus k **generator checks certify an exponentially large set** — all of $\langle S \rangle$.

Lemma 2 (the certificate characterizes equivariance — the converse). Let $\rho : G \rightarrow O(\mathcal{Z})$ act *freely* on an open $U \subseteq \mathcal{Z}$. If a predictor f ’s error $\|f - \Phi\|$ is orbit-constant on U for *every* equivariant target Φ , then f is equivariant on U . *Proof sketch (full proof and the continuity refinement in Appendix B).* Freeness makes $\Phi(\rho(h)z) := \rho(h)c$ a well-defined equivariant probe target for any c ; orbit-constancy then forces $f(z)$ and $\rho(g)^{-1}f(\rho(g)z)$ equidistant from *every* c , hence equal. $\square$ With Theorem A this is a **characterization**: orbit-constant error $\iff$ equivariance — **no non-equivariant model possesses the certificate at any size**, the impossibility a theorem, not an observation (and what the single-shot literature, forward-only, lacks).

3.2 The horizon axis, and its tightness (the central result)

Real models are only *approximately* equivariant. Let $\epsilon_{\max} = \max_i \sup_x \|E(g_i \cdot x) - \rho(g_i)E(x)\|$ be the encoder residual, δ the per-step predictor error, and let the latent Jacobian be locally diagonalized on channel j with multiplier e^{λ_j} .

Theorem B (spectral degradation — upper bound). For constants c_j depending on local geometry,

$$|\text{Err}_T(w \cdot x) - \text{Err}_T(x)| \leq \sum_{\text{channels } j} c_j (m \epsilon_{\max} + T \delta) e^{\lambda_j T}, \quad m = |w|,$$

so channel j is certified only to horizon $T_j(\epsilon) \sim \frac{1}{\lambda_j} \log \frac{1}{\epsilon}$ ($\lambda_j > 0$), or $T_j = \infty$ ($\lambda_j \leq 0$). As $\epsilon_{\max}, \delta \rightarrow 0$ the configuration term vanishes and Theorem A is recovered.

Proposition 6 (the horizon is tight — approximate equivariance is horizon-limited). Theorem B is an *upper* bound; here is a matching *lower* bound. Fix an expansive channel on which the latent map is locally linear with multiplier $a = e^\lambda$, $\lambda > 0$ (the local diagonalization of Theorem B). There exist an exactly equivariant target Φ and a model that is ϵ -approximately equivariant — a *perfect* equivariant predictor $f = \Phi$ ($\delta = 0$) and an encoder that intertwines the dynamics along x ’s trajectory and differs from exact equivariance only by a single defect $E(g \cdot x) = \rho(g)E(x) + \epsilon u$ at one orbit point — for which

$$|\text{Err}_T(g \cdot x) - \text{Err}_T(x)| = \epsilon e^{\lambda T}.$$

Proof sketch (Appendix B). Along x the model is exact, $\text{Err}_T(x) = 0$; at $g \cdot x$ the single defect propagates linearly, $f^T(E(g \cdot x)) = \rho(g)\Phi^T(Ex) + a^T \epsilon u$, while the target is $\rho(g)\Phi^T(Ex)$; subtract and use (A4): $\epsilon e^{\lambda T}$. $\square$

Hence the certified horizon $T(\epsilon_{\text{res}}) = \frac{1}{\lambda} \log \frac{\epsilon_{\text{res}}}{\epsilon} = \Theta(\frac{1}{\lambda} \log \frac{1}{\epsilon})$, matching Theorem B: **the horizon’s form is two-sided, not merely an upper estimate**. The conceptual payload is sharp:

No certificate derived from an $\epsilon > 0$ equivariance residual can promise predictability beyond $T \sim \frac{1}{\lambda} \log \frac{1}{\epsilon}$ on an expansive channel (worst case over admissible targets). Only exactness ($\epsilon = 0$) or conservation ($\lambda_j \leq 0$) yields an unbounded horizon.

This is the horizon-domain companion of Lemma 2: scale and data buy *approximate* equivariance at best (a fair augmentation baseline floors at $\epsilon \approx 10^{-4}$, never exact), and Proposition 6 amplifies that residual $e^{\lambda T}$ — a single-step tie between augmentation and equivariance **must break over horizon**.

Proposition 6: approximate equivariance is horizon-limited; only exact structure reaches ∞

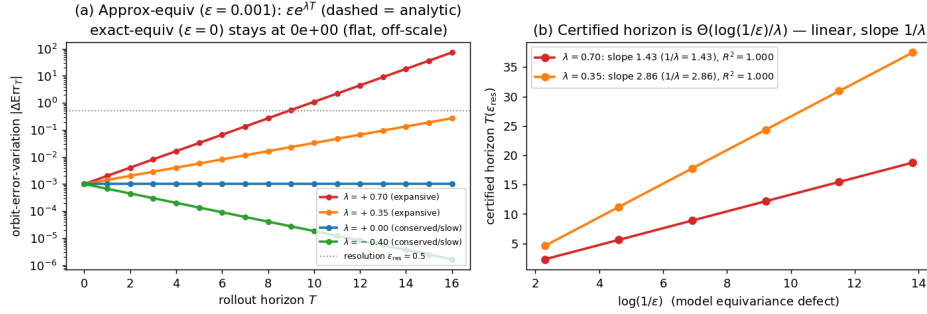


Figure 3: Proposition 6, numerically (the central claim). **Left:** orbit-error-variation of an $\epsilon=10^{-3}$ -approximately-equivariant model (markers) equals the analytic $\epsilon e^{\lambda T}$ (dashed) to relative error 10^{-14} – 10^{-13} , 3 seeds; an *exactly* equivariant model sits at the machine-precision floor; $\lambda \leq 0$ channels stay bounded (infinite horizon). **Right:** certified horizon linear in $\log(1/\epsilon)$ with slope exactly $1/\lambda$ ($R^2=1.000$) — the $\Theta(\log(1/\epsilon)/\lambda)$ law (step65).

Proposition 7 (scope — when the local spectrum certifies a *learned* model’s horizon). On a learned model the answer splits a rigorous **rate** half from an orbit-error **lift** half. Let ϕ have an ergodic invariant measure μ with $\log^+ \|D\phi\| \in L^1(\mu)$. By **Oseledets** (Oseledets 1968), $\frac{1}{t} \log \|\delta_t\| \rightarrow \lambda_1$ μ -a.e. for generic perturbations. (a) *Non-degenerate* ($\lambda_1 > 0$): $T(\epsilon) = \frac{1}{\lambda_1} \log(\epsilon_{\text{res}}/\epsilon) + o(t)$ — Theorem B’s law with λ_1 the *measurable* asymptotic rate. For a learned $\hat{\phi}$, shadowing (Pilyugin 1999) bounds only the forecast-horizon *floor* $\sim \frac{1}{\lambda_1} \log(1/\delta)$ — trajectory closeness, **not** the asymptotic exponent — so that $\hat{\phi}$ *reproduces* λ_1 is a finite-horizon continuity statement (Proposition 8), confirmed in E2. (b) *Degenerate* ($\lambda_1 \approx 0$): the log-law degenerates; the one-step spectrum carries no horizon rate. This **dichotomy is the certificate’s scope** — and it *predicts* where the certificate is vacuous (E2’s PushT-interior probe, $R^2=0.02$).

Proposition 8 (finite-horizon exponent transfer). The asymptotic exponent is only upper-semicontinuous — why shadowing cannot transfer it — but the certified horizon is **finite**, and the finite-time exponent $\lambda_1^{(T)} = \frac{1}{T} \mathbb{E} \log \|D\phi^T\|$ is locally Lipschitz over a finite orbit: a learned $\hat{\phi}$ with one-step fidelity δ recovers the staircase slope up to $O(\delta)$ plus the finite- T truncation, T -uniform under a dominated splitting (Bochi and Viana 2005). The match is **falsifiable** — the exponent must tighten as one-step error drops — confirmed in E2 (Rössler: 44% $\rightarrow$ 8%).

3.3 The Noether hinge: which channels are unbounded

Theorem B leaves the $\lambda_j \leq 0$ (unbounded-horizon) channels unidentified; the Noether hinge identifies them as the *conserved/invariant* ones.

Proposition 4 (isotypic placement). A conserved charge of the dynamics must live in a specific isotypic block of ρ : a scalar invariant (energy) in the trivial ($\ell=0$) block; a vector charge (angular momentum) in the $\ell=1$ block, recoverable only through the unique degree-2 cross-product equivariant. Placement is *forced* by representation theory (Goodman and Wallach 2009), not chosen.

Proposition 5 (conservation $\Rightarrow$ unbounded horizon). Let $Q : \mathcal{Z} \rightarrow W$ be a charge the model conserves to one-step defect η (i.e. $\|Q(fz) - Q(z)\| \leq \eta$). Then the T -step *charge-value* prediction error satisfies

$\|Q(\hat{z}_T) - Q(z_T)\| \leq T\eta$ — **linear in T** , never the chaotic $e^{\lambda T}$ — and at $\eta=0$ the charge value is certified to all horizons. (About the charge value; exact under an equivariant symplectic discretization; the converse fails — slow need not be conserved.)

Together: **the certified region is the coarse, invariant/conserved, low-composition corner** — reachable, by the converse, only with structure.

3.4 The certificate is a procedure

The certificate is *computable a priori*. Algorithm 1: (i) check (A1)–(A4) on the k generators to residual $\epsilon_{\max}$; (ii) estimate the predictor spectrum at the query latent (block-bootstrap CI, Liouville-anchored; **step78**); (iii) report whether (w, T, ϵ) lies in the certified region of Theorem B/Proposition 6, escalating conserved channels via Proposition 5. Proposition 10 gives the finite-sample rate: the bootstrap CI brackets $T_1 \in [\log(1/\epsilon)/(\hat{\lambda}_1 \pm \varepsilon_n)]$ with $n \asymp \log(1/\delta)/\varepsilon^2$ — cheap in confidence, quadratic in precision.

4. Experiments

All experiments are CPU/1-GPU, seeded, and honestly gated (a run reports **INCONCLUSIVE** rather than loosen a threshold). Proofs: **Appendix B**; reproducibility map (anonymized code, seeds, gates): **Appendix C**; the supporting suite (E1, E3–E8) is summarized below and written up in **Appendix D**.

(E2) The horizon staircase — and where structure becomes necessary. On a controlled latent with a planted spectrum the certified-horizon slope brackets the predicted $1/\lambda$ (chaotic exponent recovered to 0.4%). The law lifts to learned models of *real* chaotic dynamics: one-step models of **Lorenz**, **Hénon**, **Rössler** give staircases linear in $\log(1/\epsilon)$ ($R^2=0.975\text{--}0.995$, 3 seeds) whose slopes recover the textbook exponents to 1–12% — Proposition 8’s continuity at work, not a shadowing corollary — while the near-neutral **PushT interior** is Proposition 7’s predicted *degenerate* branch ($R^2=0.02$). At high dimension structure becomes *necessary*: on 40-D **Lorenz-96** (Lorenz 1996) (Liouville sum $\sum_j \lambda_j = -N$ recovered to 0.0%) a $\mathbb{Z}_N$ -equivariant cyclic-conv recovers the *full* spectrum ($R^2=0.98\text{--}0.99$, 3 seeds; Kaplan–Yorke ~ 27) — hence the per-channel certified horizons — while an identically-trained dense MLP **fails** ($R^2 < 0$, λ_1 inflated $\sim 3.4\times$) and a GRU-BPTT (Vlachas et al. 2020) fails the same way, its joint-state Jacobian breaking the conditional-Lyapunov condition (Hart 2024). Sweeping $N \in \{12, 20, 28, 40\}$ shows a **phase transition**: all three architectures tie through $N=28$; only the equivariant one survives $N=40$ (**step83**).

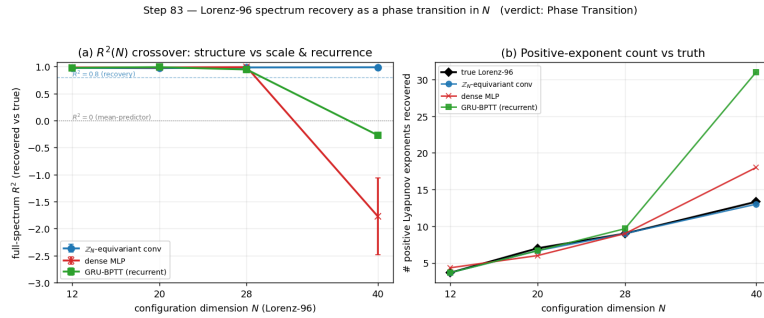


Figure 4: Full-spectrum Lyapunov R^2 vs. N : the $\mathbb{Z}_N$ -conv holds while dense MLP and GRU collapse at $N=40$ — a phase transition, not a single- N artifact (**step83**).

(E1, E3–E8: the supporting suite — Appendix D.) k generator checks certify all 2^k compositions ($\mathbb{Z}_2^6$: error $\sim 10^{-33}$ vs a baseline’s 0.59) (E1). Symmetry breaking degrades the certificate linearly — Proposition 6’s predicted crossing — and a fair augmentation baseline is horizon-limited at its $\sim 10^{-4}$ floor (E3). An $88\times$ -scaled non-equivariant model buys interpolation, never the orbit-floor (E4); on PushT contact dynamics a learned $\text{SO}(2)$ -equivariant model is orbit-flat to ratio 1.000 with no baseline in a $160\times$ sweep reaching its out-of-distribution floor — the impossibility carried by Lemma 2, not E5’s magnitude (E5). The certificate lifts

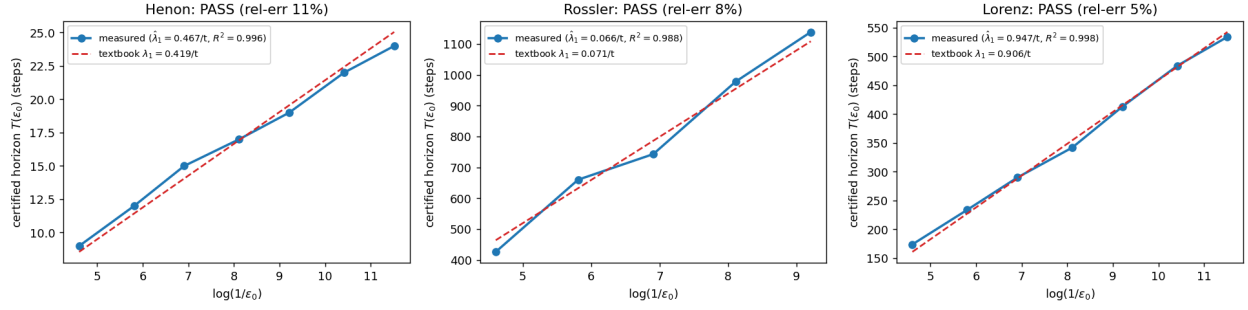


Figure 5: The certified-horizon law across a class of learned chaotic models (E2). The identical learned-model staircase on a 2D map (Hénon), a small-exponent flow (Rössler), and a large-exponent flow (Lorenz) is linear in $\log(1/\epsilon_0)$ and its slope (blue) recovers the textbook exponent (red dashed). The law is a property of chaotic dynamics, not of Lorenz; the residual bias is decomposed by Proposition 8.

to non-abelian $SO(3)$ and, via frame averaging, accuracy-neutrally to pixels (E6); becomes orbit-invariant closed-loop control (E7); and drives a noise-immune epistemic explorer — 2^7 compositions certified in 7 observations where prediction-error curiosity is lured by distractors to 1% (E8).

(E9) Both axes on one system; the certificate acts. On controlled Lorenz-96 (exact $\mathbb{Z}_N$, $\lambda_1 \approx 1.8$) an equivariant planner gives orbit-flat control (residual 8×10^{-16}) while the certified $T_1(\epsilon)$ drives an active re-observation schedule sitting on the accuracy-observation frontier *untuned* ($2/3$ seeds, asymptotic- ϵ regime; tight- ϵ honestly optimistic per Prop. 8). The pattern lifts to a pendulum ring and a double pendulum (step79–step81).

(E10) The certified horizon, made literal — and its exact regime. E2 *measures* the exponent; Theorem B *certifies* it: a computable cone/adapted-metric certificate reads a **sound, a-priori** bound on the model’s top exponent off its Jacobian field alone. On **uniformly-hyperbolic** dynamics it is *tight*: cat-map certified exponent exact on the true map, $1.17\times$ on a *learned* net (3 seeds, sound), $1.06\times$ on a nonlinear Anosov perturbation. On **non-uniformly-hyperbolic** Hénon a single metric is provably limited — sound but $\sim 3\times$ conservative — and the certificate’s own cone-margin diagnostic turns negative: it **abstains** rather than over-claim. A tight a-priori horizon is achievable *exactly* in the uniformly-hyperbolic regime, and the certificate **self-diagnoses** its regime (step82).

(E11) On a recognized control benchmark — honest INCONCLUSIVE. On Acrobot-v1 ($\lambda_1 = 0.094$) the certified horizon *tracks* the measured one (ratio $0.42 \rightarrow 0.93$ as ϵ coarsens) and *binds*: planning past $\sim T_1$ fails outright, and capping depth at T_1 solves the task (67–100%) — but the return-optimum sits at $\sim T_1/2$, so two pre-registered no-tuning rules are **INCONCLUSIVE** against the best-tuned blind depth (step84).

(E12) The certificate’s trustworthiness changes a budgeted decision — a priori. On 40-D Lorenz-96 an agent schedules sparse re-observations of a chaotic forecast under a fixed sensing budget. The forecaster is held fixed (the equivariant model; both models are identically trained and forecast comparably, one-step relMSE $\sim 10^{-5}$) and *only* the certificate timing re-observation varies: the equivariant certificate yields 8–16% aggregate violation at the knee budget vs. 61–65% for the dense certificate, whose λ_1 is inflated $\sim 3.4\times$ and which therefore over-observes and starves the budget (margins $+0.45$ – $+0.57$, $3/3$; **Proposition 9** makes the cost a law — $c\times$ inflation needs $c\times$ budget; measured catch-up 2.7 – $3.5\times$). Two controls scope the claim: a certificate-free adaptive scheduler matches only after $\sim 3\times$ the budget (the certificate is an *a-priori* warm-start), and a *recalibrated* dense certificate closes the gap ($3/3$) but only by **spending a calibration set** — the equivariant certificate is correct from the Jacobian with **zero rollout data**. Standard UQ baselines (a 4-model ensemble, a 10-rollout conformal quantile) calibrate tighter by spending training or truth access the certificate does not: **no method dominates the accuracy–cost Pareto; the certificate is its only a-priori point** (step90). The contrast replicates **conditionally** — the condition being the mechanism — on a pendulum ring ($N=24$, 5 seeds): the gap appears exactly where the dense λ_1 inflates ($2/5$) and

vanishes where it is incidentally faithful — chance vs. structure. (A budget *allocation* across a chaoticity ensemble did *not* win; reported honestly.) (step85–step88.)

(E13) The certificate audits a public pretrained world model — training-free, and maps onto the paper’s own scope theory. We rebuild the latent slices of the official TD-MPC2 checkpoints and run the *unchanged* machinery on the policy-prior loop $g(z) = d(z, \tanh \mu_\pi(z))$ — no training, no environment access on the certified side — across **5 tasks $\times$ 3 official seeds (15 loops)**. The result is a scope map tracking the theory cell-by-cell. Where the loop is **strongly expansive** ($\lambda_1=0.25\text{--}0.30$: walker-walk 3/3, cheetah-run seed 3) the coarse-resolution certificate is **calibrated** — ratio measured/certified 0.83–1.02 (walker 0.94/0.95/1.02) (tight- ϵ optimistic as everywhere; Prop. 8). As expansion **weakens** the certificate turns optimistic (cheetah 0.43/0.50; hopper-hop 0.13/0.38 at $\lambda_1=0.05\text{--}0.09$): model bias outpaces Lyapunov amplification — the degeneracy direction Proposition 7 flags. Where the loop **contracts** (6/15: acrobot 3/3, finger-spin 2/3, hopper seed 1) the certificate **abstains, correctly in both sub-cases**: finger-spin’s stable loops genuinely do not diverge (nothing to certify; 15–19/20 starts censored), while acrobot’s and hopper-1’s residual divergence is bias-driven — outside a Lyapunov certificate’s jurisdiction. SimNorm’s structural zero-directions appear as a strongly-negative band — reported, not hidden; the certified scope is the prior loop, not the planner. A Jacobian certificate of a public zoo’s *learned latent map*, cross-validated against true-environment divergence and stratified by the certificate’s own scope theory, is new (nearest prior: true-environment Lyapunov under RL policies, arXiv:2410.10674). (step89.) A second, architecturally disjoint family lands on the same map: the official **LeWM** checkpoint (pixel-input ViT+transformer JEPa, loaded bit-faithfully into the authors’ own code) has $\lambda_1=0.001$ with CI straddling zero — the certificate **abstains**, and the observed 1–2-step divergence is pure one-step bias, the same sub-case as acrobot. Two families, one read-out, one taxonomy (step91).

(E14) Scale does not rescue trustworthiness (step92). Across the official TD-MPC2 *multitask* ladder (mt30, 1M–317M, same walker-walk task, one official checkpoint per size) the loop’s regime flips **non-monotonically** with scale — contracting at 1M and 48M, expansive at 5M/19M/317M — and calibration scatters (0.37/1.87/1.16 at $\epsilon=0.2$ where expansive; mt80 likewise mixed), **no size matching the single-task** 0.94–1.02. One checkpoint per cell — a descriptive scope-map extension, unambiguous in direction: *trust in a rollout is a property of the loop’s dynamics, not of parameter count — scale buys interpolation, not a calibrated horizon*.

Scale does not buy a calibrated horizon — official TD-MPC2 multitask ladder, walker-walk (1 ckpt/cell)

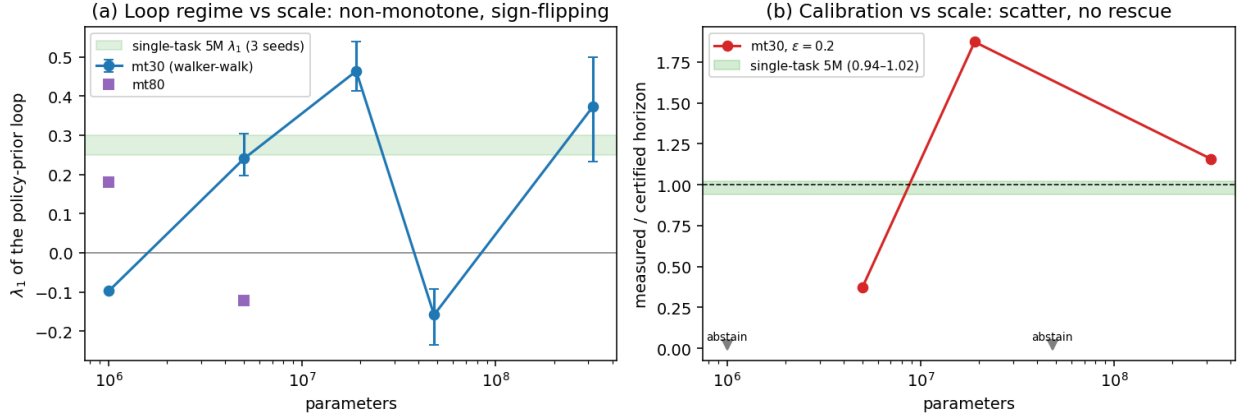


Figure 6: Scale does not buy a calibrated horizon (E14). **(a)** λ_1 of the walker-walk policy-prior loop across the official multitask ladder: sign-flipping, non-monotone (contracting at 1M and 48M). **(b)** Calibration at $\epsilon=0.2$: scatter across sizes; no multitask scale reaches the single-task 5M band (0.94–1.02, green).

(E15) The published certificate prices a deployed monitor, out-of-sample (step94). The scope law’s *positive* instance: a **sensor-only monitor** watches cheetah-run under its nominal policy, reads the expensive sensor every k steps, and between reads forecasts with the certified loop g itself (no action telemetry),

flagging at a read iff relative error exceeds $\theta=0.2$. Certificate numbers are **loaded from the E13 artifact** — issued before this experiment existed — and the gates were frozen before seeds 1–2 ever ran. The in-situ staleness clock replicates the published map **cell-by-cell**: in-situ-vs-bench ratio 0.43 vs 0.43 and 0.50 vs 0.50 on the out-of-sample cells (optimism *predicted*), 0.67 vs 0.83 on the calibrated cell — its $[2/3, 3/2]$ check landing 7×10^{-4} below the edge on an integer-valued median, recorded **at-the-edge, not rounded up**. A frozen-actuator fault is then detected at the certificate-derived cadence with recall 1.00 on 3/3 seeds (median delay $\leq k_{\text{op}}$ on 2/3). Proposition 11 formalizes both sides: an aligned decision transfers certificate value with **zero new estimation** (clause i); step93’s dilution is a resolution mismatch, $H(\theta^*) \approx 2$ vs $H(0.2) \approx 6$ (clause ii). Honest notes: on walker the deterministic prior is regime-bimodal and clock replication breaks (0.32–0.47 vs 0.94–1.02, 0/3; fault recall intact) — a monitor presumes a nominal regime, **Proposition 7 load-bearing in deployment**.

What does structure buy, if the read-out audits any smooth model? The law applies to any C^1 latent map — hence E13’s audits of non-equivariant models. What it cannot supply there is *trust in the number itself*: a dense spectrum can be silently wrong while predictions stay good (E2), so a generic certificate must be cross-validated against held-out divergence — the per-model check E13 performs. Structure removes that requirement where it holds: faithfulness (E2) makes the certificate a-priori trustworthy with zero calibration data (E12) — and exclusively so (Lemma 2). **The audit is universal; the a-priori guarantee is structure’s.**

5. Related work

Single-shot certified equivariance is our $T=1$ slice. Equivariant classifiers carry *orbit-constant margins* giving uniform adversarial certificates (*Orbit-Constant Margins for Equivariant Networks* 2025); invariance-aware randomized smoothing builds orbit-based certificates (*Invariance-Aware Randomized Smoothing Certificates* 2022); equivariantized conformal prediction group-averages the nonconformity score (Bousias et al. 2026). **All are single-shot, forward-only, and have no horizon, spectrum, or conservation axis** — exactly the $T=1$, ϵ -independent slice of our certificate; our contribution is the multi-step stratification (Theorem B, tight by Proposition 6), the converse (Lemma 2), and the Noether bridge (Proposition 5).

Learned conservation laws. Noether Networks (Alet et al. 2021) and Noether’s Razor (*Noether’s Razor* 2024) *learn* conserved quantities to improve average prediction; we *certify* — Proposition 5 turns a conserved charge into an a-priori long-horizon guarantee, Proposition 4 forces its isotopic placement. Guarantee versus average accuracy.

Jacobian-regularized world models. (*Towards Unraveling and Improving Generalization in World Models* 2025) penalizes the latent-transition Jacobian to damp rollout error — a heuristic for robustness. Theorem B is the provable version of the intuition: read a per-channel certified horizon off the spectrum instead of regularizing toward stability; Proposition 6 prices its decay under approximate symmetry.

Equivariant predictors and latent-geometry priors. Cyclic/unitary predictors (BRO-JEPA, UWM-JEPA) report strong zero-shot transfer; Theorem A explains *why* and Lemma 1 quantifies *how far* (the generated monoid). LeJEPA-style isotropic latent priors target a *distributional* property; ours is a first-order, per-situation guarantee that predicts when a data-discovered anisotropy fails off the orbit.

Predictability horizons. The $T(\epsilon) \sim \log(1/\epsilon)/\lambda$ law is classical (Lyapunov; NWP); the local-to-asymptotic link is Oseledets (Oseledets 1968); shadowing (Pilyugin 1999) bounds a perturbed model’s horizon *floor*, not its exponent. We (i) prove the law *tight* for a learned latent world model and measure it on a class of genuinely chaotic dynamics (1–12%, E2), rigorous via Proposition 8 where shadowing fails; (ii) characterize *when* it lifts (Proposition 7); (iii) tie the unbounded-horizon subspace to the invariant one (Noether); (iv) prove the converse (Lemma 2). Spectrum recovery from a learned model is known *conditionally* on contracting modes (Hart 2024) — why the recurrent baseline fails at high N ; the closest concurrent guarantees (Conradie et al. 2026; Lillemark et al. 2026; Mo 2026) each miss the equivariance-certified-horizon intersection.

Concurrent work (2026, briefly). The nearest neighbors are symmetry-protected *neutral* Lyapunov modes (Mo, arXiv:2605.03338 — constrains the spectrum’s kernel where we certify the horizon; for our discrete $\mathbb{Z}_N$ systems their bound is vacuously zero), conformal rollout bounds (Geng et al., arXiv:2512.08991 — statistical and rollout-hungry where ours is a-priori and training-free), and flow-equivariant world models ((Lillemark et al. 2026) — an exactness/closure property, not a quantitative horizon). A Jacobian certificate of a *public* model’s latent map, cross-validated against true-environment divergence, is to our knowledge new. **Appendix E** gives the full sweep (through 2026-06) and the per-work distinctions.

6. Limitations

- **Exactness needs a genuine dynamical symmetry (A3).** Exact where G is a symmetry of the dynamics, *gracefully approximate* elsewhere — the degradation two-sidedly bounded (Theorem B, Proposition 6) and measured (E3).
- **Tight in form, not in prefactor.** The $\log(1/\epsilon)/\lambda$ *form* is tight; the constants c_j are not estimated, and the isotypic refinement is asserted, not separately measured.
- **The Noether hinge: forward direction proved, hypotheses measured, emergence open.** Proposition 5 assumes a G -invariant Hamiltonian latent flow (the defect η exact only under an equivariant symplectic discretization), and the converse fails (slow $\nRightarrow$ conserved). The hinge is moreover validated on *constructed* equivariant teachers — the real-ish experiments (E5 PushT, E6 SO(3)) validate *flatness*, not the hinge’s *emergence* in a learned model of un-constructed dynamics; that remains open.
- **The horizon law lifts to learned chaotic models and is vacuous on near-neutral dynamics (scope, not a bug).** Validated on a synthetic spectrum, a *class* of low-D chaotic systems (exponents to 1–12%), and a 40-D learned model ($R^2=0.98$ –0.99); informative iff $\lambda_1 > 0$ — the PushT interior is the predicted degenerate branch ($R^2 \approx 0.02$, Prop. 7(b)). Caveats: the lift rests on Proposition 8 (not shadowing), and we verify C^1 -closeness only via one-step L^2 error — a real gap in high N (one-step-accurate dense and recurrent models mis-estimate the 40-D spectrum; structure closes it); the real-dynamics evidence is ODEs/maps, not a contact simulator.
- **Demonstrated downstream value is budgeted efficiency, not safety.** A separate test of the certified horizon as a *catastrophe-avoidance* cadence was **inconclusive** at our scale (escape was re-observation-interval-invariant — control-quality-limited); the certificate’s *necessity* for safety is open.
- **Where decision value concentrates — and where it dilutes.** Closing the loop on the real TD-MPC2 agent (faithful MPPI replica; cadence-1 anchor return 994 vs the official 977–983 band), return degrades from replan cadence $k=2$ — well inside $T_1(0.2) \approx 5.4$ –6.4; the control-relevant resolution is finer and sits in the certificate’s known tight- ϵ optimistic regime (**step93**). With E11, the honest scope law: value concentrates where the decided quantity IS the certified quantity (E12’s re-observation win, E15’s deployed monitor) and dilutes behind a task-level map (return, gait quality). **Proposition 11 makes the law a theorem:** an aligned decision inherits certificate value at calibration cost alone (zero regret at $c=1$); a task-mapped decision pays an irreducible mis-resolution penalty $|\log(\epsilon/\theta^*)|/\lambda_1$, and the tolerance θ^* is the task’s to give, not the certificate’s — the boundary of a-priori decision value, stated rather than promised past.
- **Scale and modality.** All experiments are 1–2-GPU. The certificate’s exact flatness transfers across modalities (state, SO(3) point clouds, pixels), but absolute multi-step accuracy on raw pixels is poor for *every* architecture at this scale (the anti-collapse JEPA latent, not equivariance); a strong few-step pixel predictor is an architecture-agnostic open problem orthogonal to the certificate.

7. Conclusion

An equivariant world model can certify, a priori and without retraining, which situations it will handle and **for how many steps** — and the horizon is tight. The certified region is the coarse-invariant-slow-low-

composition corner; its boundary is the predictor spectrum; its unbounded edge is the conserved/invariant subspace; the whole region is reachable only with structure. Because the horizon is *faithful* it is *actionable* — budgeting sensing on a chaotic forecast (E12), auditing public world models (E13–E14), pricing a deployed monitor (E15). *Scale buys interpolation; structure buys a certified horizon.*

Appendix D — supporting experiments (full write-ups)

(E1) The configuration axis is exponential. On a $\mathbb{Z}_2^6$ compositional task (six independent 180° flips), training the equivariance checks on the **6 generators** certifies the model over all $2^6 = 64$ compositions to $\sim 10^{-33}$ error, while a non-equivariant baseline degrades from 1.6×10^{-5} on the generators to 0.59 on held-out compositions. Six checks, sixty-four guarantees (Lemma 1).

(E3) Approximate symmetry validates Proposition 6. Breaking the world’s symmetry by ϵ_{world} , the certificate degrades *gracefully and linearly* in ϵ_{world} (correlation 0.88–0.98 across seeds) up to a measured threshold $\epsilon_{\text{world}} \approx 0.01$ –0.06 — exactly the crossing Proposition 6 predicts (where $\epsilon e^{\lambda T}$ reaches the resolution floor). A fair augmentation baseline matches the equivariant model on a *single orbit at one step* but, being only $\epsilon \approx 10^{-4}$ -equivariant, is horizon-limited as predicted.

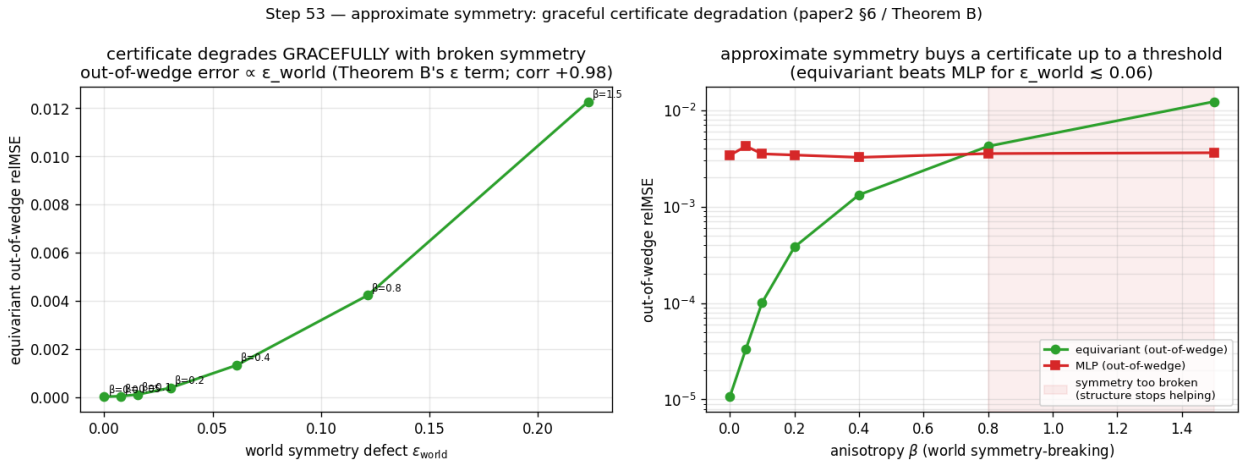


Figure 7: Approximate symmetry (E3). **Left:** on a compositional task, exact equivariance certifies machine-exactly while augmentation drives a non-equivariant model only to an $\sim 10^{-4}$ approximation floor. **Right:** breaking the world’s symmetry by ϵ_{world} degrades the certificate gracefully and *linearly* (the slope Proposition 6 predicts) up to a measured threshold.

(E4) Structure vs. scale. The equivariant model is orbit-flat (ratio 1.1–1.2); an $88\times$ -scaled non-equivariant model buys in-distribution interpolation (31 – $166\times$ better in-wedge, *beating* the equivariant model there) but its out-of-wedge error stays 10 – $155\times$ above the equivariant floor. Scale buys interpolation; it never buys the certificate (Lemma 2).

(E5) The certificate on real contact dynamics. On PushT — a pymunk physics engine whose square arena makes $\text{SO}(2)$ scene rotation an exact dynamical symmetry — a *learned* $\text{SO}(2)$ -equivariant world model has 10-step rollout error **exactly flat over the orbit** (ratio 1.00, equivariance residual $\sim 10^{-7}$) and is competitive in-distribution (0.13–0.15 vs. the best baseline 0.14–0.19); **no non-equivariant baseline across a $160\times$ parameter sweep** ($1.7\text{k} \rightarrow 272\text{k}$) **reaches the equivariant floor out of distribution.** We are careful about the *size* of that gap: per baseline it is $\sim 3.9\times$ at the *smallest* model and 2.1 – $2.6\times$ at the *largest*, i.e. it does **not** grow with scale — the honest control is the strong-baseline $\sim 2\times$. The *force* of E5 is the exactness of the flatness (ratio 1.000 vs. the baselines’ 2 – $3\times$ drift) and the a-priori certificate, not

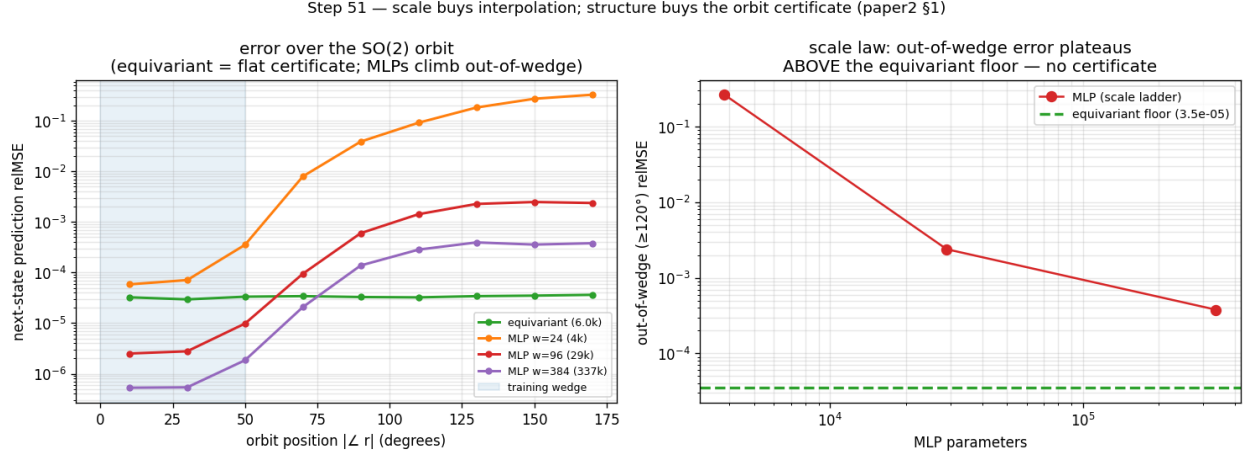


Figure 8: Structure versus scale (E4). The equivariant model is orbit-flat (a certificate); an $88\times$ -scaled non-equivariant model buys *in-distribution* interpolation (it beats the equivariant model inside the training wedge) but its error climbs $10\text{--}155\times$ above the equivariant floor *out* of the wedge — scale buys interpolation, not the certificate.

the OOD-error magnitude (which on one $SO(2)$ task is modest); the “scale cannot buy it” claim is carried by Lemma 2 / §3.3, not by E5’s gap. The dynamics were not authored by us. (Honest gate note: the run’s load-bearing *flat* and *floor* sub-gates pass on all 3 seeds; the auxiliary “MLP error climbs with horizon faster than equivariant” sub-gate is met on 1/3 seeds and reported **INCONCLUSIVE** on the others.)

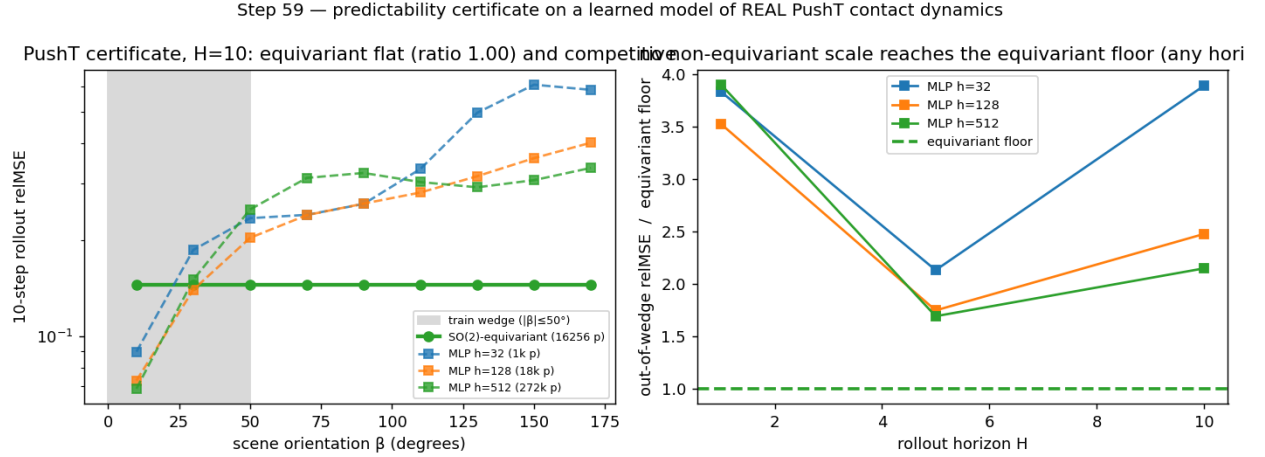


Figure 9: The certificate on real contact dynamics (E5). On PushT (a physics engine we did not author), a *learned* $SO(2)$ -equivariant world model’s multi-step rollout error is flat over the orbit of scene orientations to the floating-point floor, while a $160\times$ parameter sweep of non-equivariant baselines never reaches the equivariant floor out of distribution.

(E6) The certificate is not $SO(2)$ -specific, and transfers to pixels at no accuracy cost. It lifts to non-abelian $SO(3)$ on 3D point clouds (learned equivariant rollout exactly flat, ratio 1.000, with a $7.4\times$ -smaller model than the baseline). On raw rendered pixels (PushT, exact C_4), **frame averaging** (Puny et al. 2022) — a plain CNN/MLP made exactly C_4 -equivariant by a Reynolds average over grid rotations — keeps the exact orbit-flat certificate (ratio 1.000) while being **accuracy-neutral**: it matches or beats an unconstrained CNN on a collapse-robust metric (fraction-of-variance-unexplained ratio 0.68–1.07, mean 0.84, 3 seeds), with a healthier latent and a horizon-stable rollout where the steerable baseline diverges. The prior

costs nothing *relative to* the unconstrained CNN — though both remain limited in absolute terms (neither beats predict-the-mean at this scale; §6).

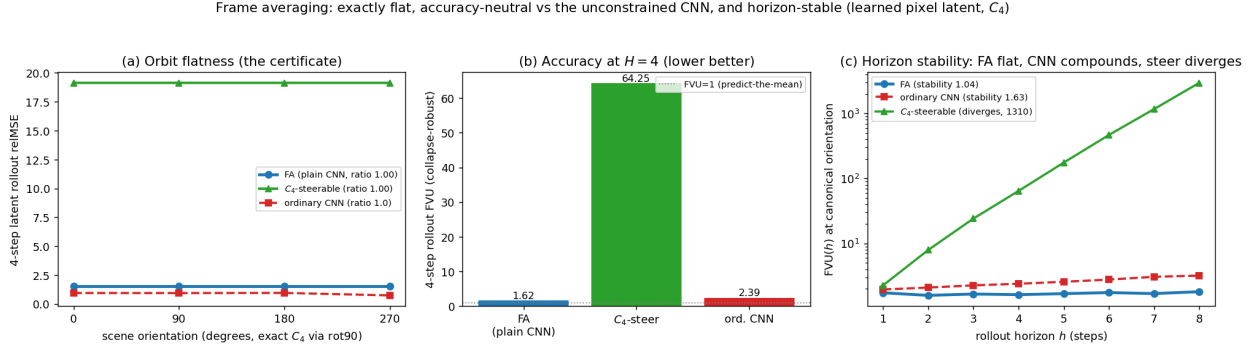


Figure 10: The certificate on raw pixels at no accuracy cost (E6). On rendered PushT frames (exact C_4), frame averaging makes a plain CNN/MLP exactly C_4 -equivariant: the latent rollout is flat over the orbit (left, the certificate), it matches or beats an unconstrained CNN on a collapse-robust accuracy metric (centre), and its rollout error stays bounded over horizon while the steerable baseline diverges (right, log scale).

(E7) The certificate becomes task competence. Run through a G -equivariant planner (A5), the prediction certificate becomes an *orbit-invariant control* certificate: closed-loop pose error is flat over the orbit to the floating-point floor (ratio 1.000, 3 seeds) while a $4.3\times$ -larger non-equivariant controller degrades out of the training wedge. The flatness is exact because (A5) makes the planner G -equivariant and the planning cost G -invariant, so the closed-loop trajectory inherits the encoder’s orbit-equivariance — Theorem A under the closed-loop clause.

(E8) The certificate is an epistemic drive: certificate-driven active inference. Because the certified region is *computable* (Algorithm 1), an agent can act to **expand** it — a provable alternative to curiosity’s “reduce prediction error”. On a $\mathbb{Z}_2^7$ compositional world whose action space mixes 7 true generators with noisy high-error distractors, an explorer that maximizes certified-region growth certifies all $2^7=128$ compositions in exactly 7 observations (the generator basis — each generator unlocks an exponential swath, Lemma 1), whereas an error-curiosity explorer is lured by the irreducible-noise distractors and certifies only 1% at the same budget (3 seeds; random $\sim 2\%$). So “expand your certified region” is a **noise-immune** epistemic drive that beats prediction-error curiosity — the classic *noisy-TV* failure mode. *Honest scope:* against *raw-error* curiosity (a sophisticated information-gain agent would also avoid aleatoric noise); the certificate supplies that immunity **for free and provably**, no noise model — a toy demonstration of the principle, not a benchmark.

(E2 supplementary figures.)

Appendix E — concurrent and recent work (full sweep through 2026-06)

Mo (arXiv:2605.03338) proves symmetry-protected *neutral* Lyapunov modes for continuously-equivariant fields ($\geq \dim(G/H)$ zero exponents along the group orbit) — complementary to ours: it constrains the spectrum’s *kernel*, while we certify the *horizon* stratified by the whole spectrum (for our discrete $\mathbb{Z}_N$ systems $\dim(G/H)=0$, so their lower bound is zero there — the two results constrain disjoint parts of the spectrum: they the kernel, we the horizon). Geng et al. (arXiv:2512.08991) bound world-model rollout deviation *conformally* for closed-loop verification — statistical and rollout-hungry where ours is a-priori and training-free; the same trade-off separates us from reachability-based MBRL safety (UPSi, arXiv:2604.26836) and where-to-trust heuristics (arXiv:2606.01363). Pretrained world models have been probed *semantically* (arXiv:2603.21546) and benchmarked by sample rollouts (WorldBench, arXiv:2601.21282; WorldArena, arXiv:2602.08971); a Jacobian certificate of a public model’s latent map, cross-validated against true-environment divergence, is to our knowledge new — the nearest priors being latent-space stability analyses of *self-trained* autoencoders (Özalp & Magri, arXiv:2410.00480) and Lyapunov-regularized DreamerV3

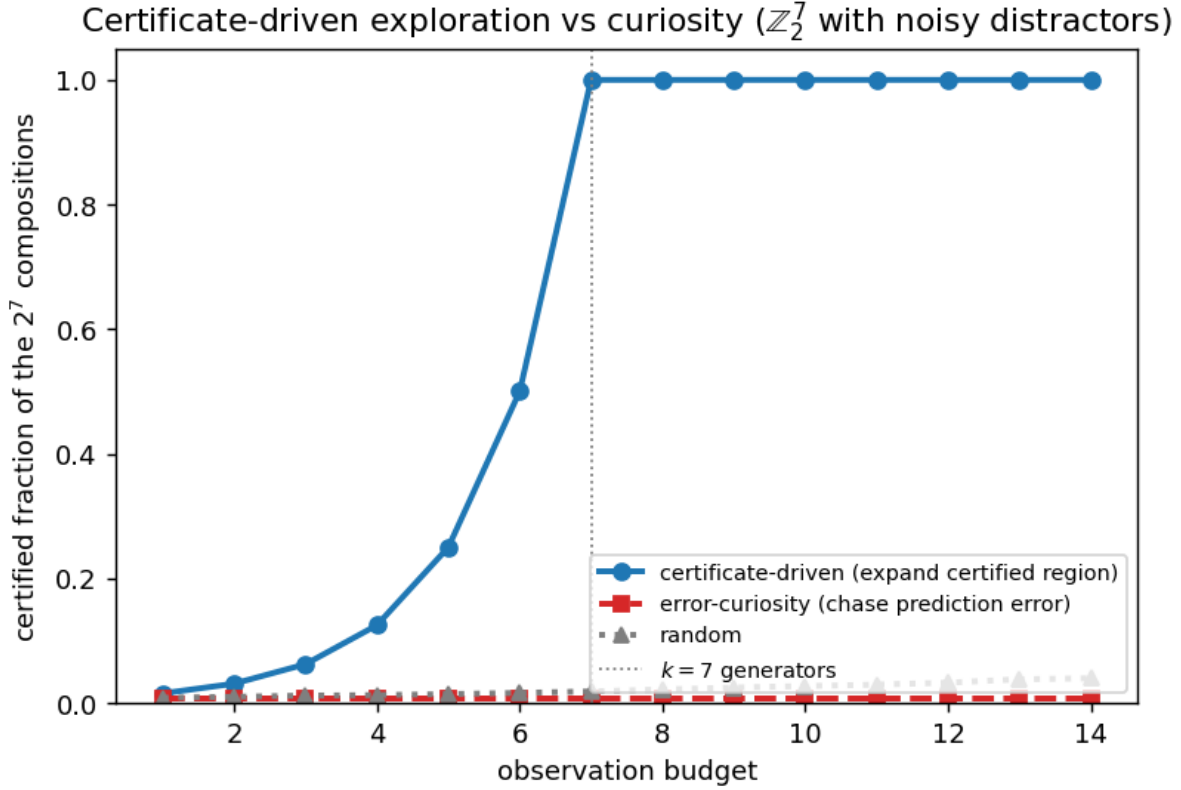


Figure 11: Certificate-driven active inference (E8). On a $\mathbb{Z}_2^7$ compositional world with noisy distractor actions, an explorer that maximizes the *certified region* certifies all 128 compositions in exactly 7 observations (the generator basis), while a prediction-error-curiosity explorer is lured by the high-error distractors and certifies almost nothing at the same budget.

The certified horizon law on a learned model of REAL chaotic dynamics (Lorenz)

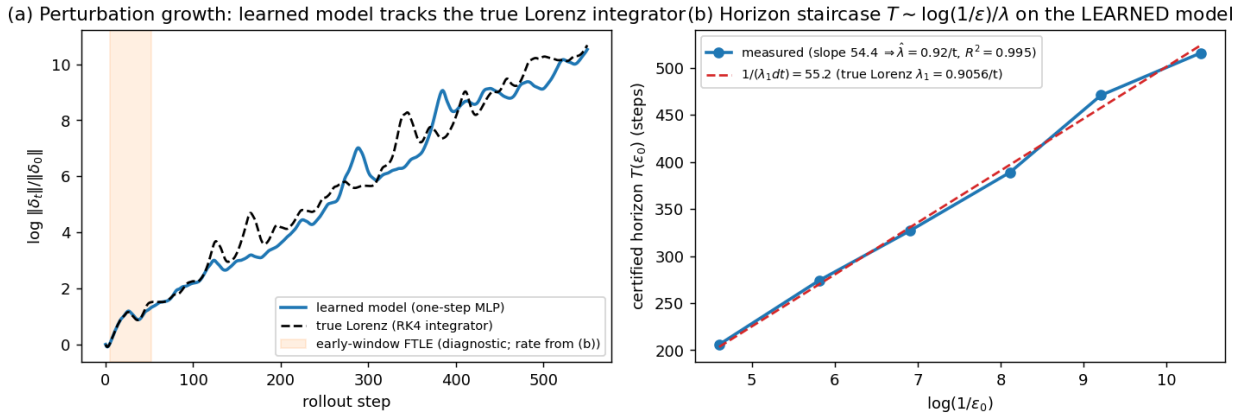


Figure 12: The horizon staircase on a learned model of real chaotic dynamics (E2, Lorenz). **Left:** the learned one-step model’s perturbation growth tracks the true Lorenz integrator over 550 steps. **Right:** the certified horizon $T(\epsilon_0)$ on the *learned* model is linear in $\log(1/\epsilon_0)$ ($R^2=0.995$), and the measured slope sits on the prediction $1/(\lambda_1 dt)$ from the textbook Lorenz exponent — Theorem B’s law lifted to a learned model of a genuinely chaotic system (Proposition 7(a)).

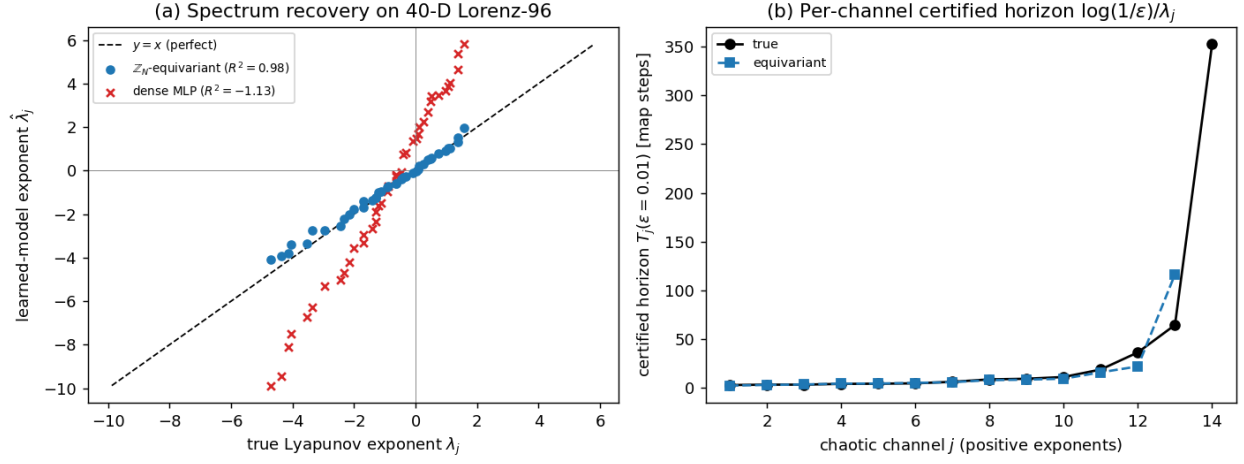


Figure 13: High-dimensional spectral horizon (E2, 40-D Lorenz-96). **Left:** recovered vs. true Lyapunov exponent for all 40 channels — the $\mathbb{Z}_N$ -equivariant cyclic-conv (blue) lies on $y = x$ ($R^2=0.98$); the dense MLP (red $\times$) is scattered far off ($R^2 = -1.1$). **Right:** per-channel certified horizon $\log(1/\epsilon)/\lambda_j$: the equivariant model tracks the truth across the spectrum. Structure recovers the high-dimensional spectral horizon a dense model of equal data cannot.

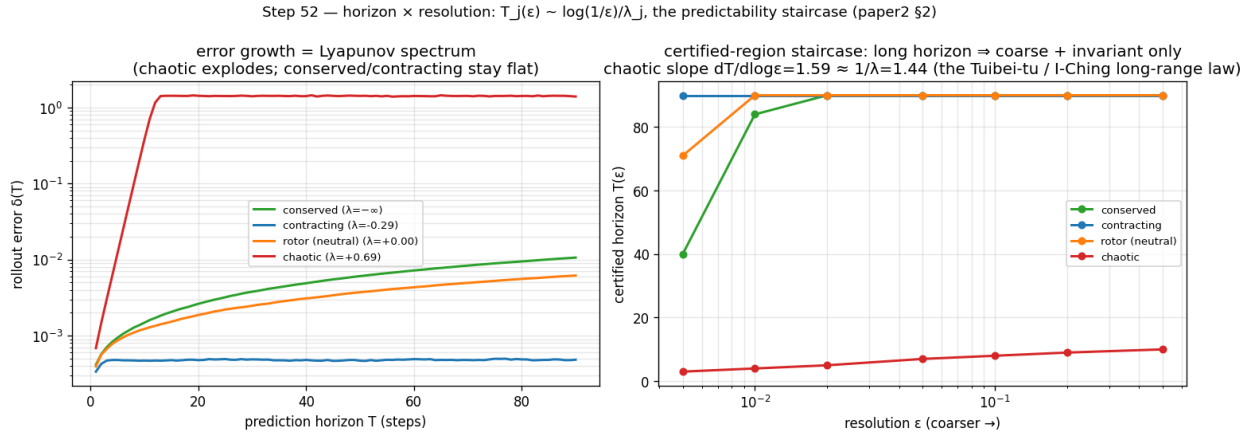


Figure 14: The horizon $\times$ resolution staircase on a controlled spectrum (E2). The measured certified horizon per channel as the demanded resolution ϵ sharpens, recovering $T_j(\epsilon) \sim \log(1/\epsilon)/\lambda_j$ (slope 1.3–1.6): chaotic channels are certified to a few steps, slow/invariant channels to dozens.

policies (arXiv:2410.10674). Flow-equivariant world models ((Lillemark et al. 2026), now ICML 2026; antecedent FERNN, arXiv:2507.14793) preserve equivariance over arbitrarily long rollouts — an exactness/closure property, not a quantitative horizon. Inductive-bias studies (arXiv:2602.06923) and position papers calling for verified world models (arXiv:2602.23997) support the framing; PDEder (arXiv:2603.22655) *suppresses* latent Lyapunov exponents where we *certify* them; two-sided calibration bounds under equivariance (arXiv:2510.21691) concern calibration error, not horizon; and the data-assimilation literature’s observation-frequency-vs-Lyapunov-time rule (e.g. Bocquet et al. 2026) is the classical neighbor of our sensing-budget law (Proposition 9).

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